NETWORKING PROJECT REPORT

# Enterprise Network Design and Implementation

## using Cisco Packet Tracer

Prepared by

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| | | |
|---|---|---|
| **3** Buildings | **6** Departments | **2** ISP links |
| **4** Routers | **2** Multilayer switches | **6** Access switches |

Routing: OSPF | DHCP relay | NAT overload | DNS and HTTP servers | Backup ISP path

# Abstract

This report describes the design and implementation of a complete company network in Cisco Packet Tracer. The company has three buildings and six departments, and every department has its own /26 subnet. All devices receive their addresses automatically from one central DHCP server through DHCP relay on the multilayer switches.

Inside the company, OSPF area 0 routes traffic between all subnets. NAT overload on two core routers gives internet access through two ISPs, with Core1 as the primary path and Core2 as the backup. DNS and HTTP servers in the Server Room provide name resolution and a company web page. The report explains the addressing plan, device configuration, server setup and the tests used to verify the network.

**Technologies used:** subnetting, DHCP and DHCP relay, OSPF, NAT/PAT, floating static route, DNS, HTTP, wired and wireless clients.

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# 1. Introduction

This project is a complete company network built in Cisco Packet Tracer. The company has **three buildings** and **six departments**: Sales & Marketing, HR & Logistics, Finance & Account, Inter-Relation, ICT Engineer and the Server Room.

Every department has its own subnet, and all devices in it get their IP address automatically from a central DHCP server. The network uses two ISP connections for internet, two core routers, two multilayer switches and six access switches. OSPF is used for routing inside the company, and NAT is used on the core routers to reach the internet.

## 1.1 Objectives

- Design a scalable, department-wise network with separate subnets.
- Automate IP assignment using a central DHCP server with DHCP relay.
- Provide name resolution (DNS) and a web service (HTTP) to all departments.
- Use OSPF for dynamic routing between all subnets.
- Provide internet access through NAT, with a backup ISP path for redundancy.
- Support both wired and wireless end devices.

## 1.2 Main points of the project

| Feature | Details |
|---|---|
| Subnetting | Separate /26 subnet for every department |
| Central services | DHCP, DNS and HTTP servers in the Server Room |
| DHCP relay | `ip helper-address` on the multilayer switches |
| Routing | OSPF area 0, default route from the core routers towards the ISPs |
| Internet access | NAT overload (PAT) on both core routers |
| Redundancy | Core1/ISP1 is primary, Core2/ISP2 is backup |
| End devices | Wired (PC, printer, laptop) and wireless (smartphone, tablet, laptop) through access points |

## 1.3 Device inventory

| Device type | Model | Qty | Names |
|---|---|---|---|
| Router | Cisco 2911 | 4 | ISP1, ISP2, Core1, Core2 |
| Multilayer switch | Cisco 3560-24PS | 2 | MLS1, MLS2 |
| Access switch | Cisco 2960-24TT | 6 | SW-SALES, SW-HR, SW-FINANCE, SW-INTERREL, SW-ICT, SW-SERVER |
| Server | Server-PT | 3 | Server0 (DHCP), Server1 (DNS), Server2 (HTTP) |
| End devices | PC, laptop, printer, tablet, smartphone | Per department | PC, printer, 2 laptops, tablet, smartphone |
| Access point | Access Point-PT | 5 | One per user department |

# 2. Network Topology

Each department has one 2960 access switch with a PC, a printer, a laptop and an access point. The access point connects a smartphone, a tablet and another laptop wirelessly. The access switches connect upwards to the multilayer switches, which connect to the core routers and then to the ISPs.

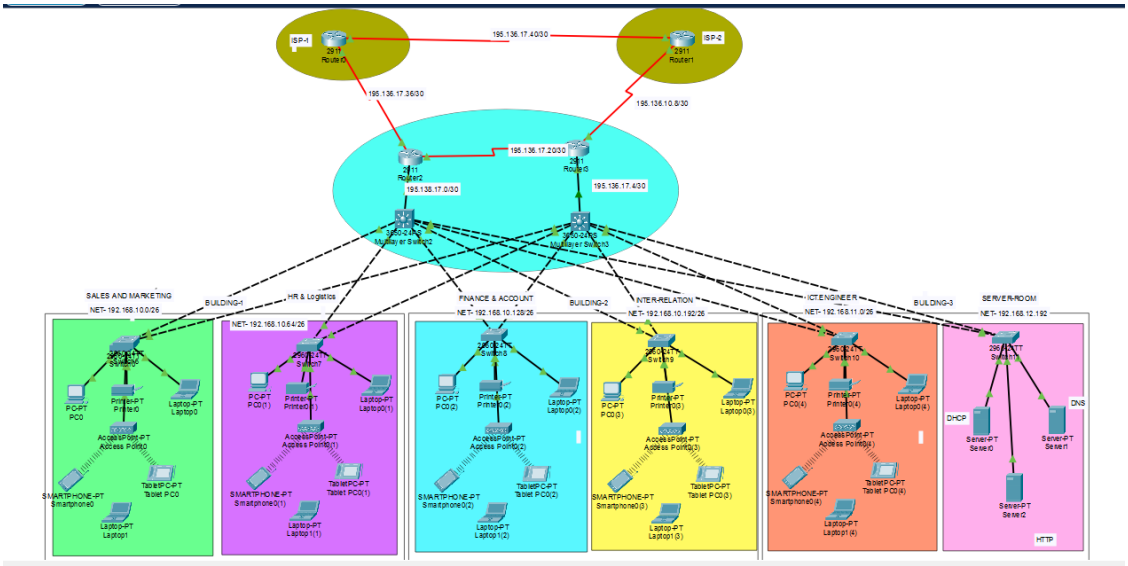



**Figure 1:** Full network topology showing ISP1, ISP2, the core routers, the multilayer switches and the departments of Building-1, Building-2 and Building-3

## 2.1 IP addressing plan

| Department | Building | Network | Subnet mask | Gateway | DHCP pool starts at |
|---|---|---|---|---|---|
| Sales & Marketing | Building-1 | 192.168.10.0/26 | 255.255.255.192 | 192.168.10.1 | 192.168.10.10 |
| HR & Logistics | Building-1 | 192.168.10.64/26 | 255.255.255.192 | 192.168.10.65 | 192.168.10.74 |
| Finance & Account | Building-2 | 192.168.10.128/26 | 255.255.255.192 | 192.168.10.129 | 192.168.10.138 |
| Inter-Relation | Building-2 | 192.168.10.192/26 | 255.255.255.192 | 192.168.10.193 | 192.168.10.202 |
| ICT Engineer | Building-3 | 192.168.11.0/26 | 255.255.255.192 | 192.168.11.1 | 192.168.11.10 |
| Server Room | Building-3 | 192.168.12.192/29 | 255.255.255.248 | 192.168.12.193 | Static IPs |

**Server Room static addresses**

| Server | Role | IP address | Subnet mask | Gateway |
|---|---|---|---|---|
| Server0 | DHCP | 192.168.12.194 | 255.255.255.248 | 192.168.12.193 |
| Server2 | HTTP (web) | 192.168.12.196 | 255.255.255.248 | 192.168.12.193 |
| Server1 | DNS | 192.168.12.197 | 255.255.255.248 | 192.168.12.193 |

## 2.2 Links between network devices

All links between routers and multilayer switches are point-to-point /30 networks (255.255.255.252).

| From | To | Network | From IP | To IP |
|---|---|---|---|---|
| MLS1 Gi0/1 | Core1 Gi0/0 | 10.0.1.0/30 | 10.0.1.1 | 10.0.1.2 |
| MLS1 Gi0/2 | Core2 Gi0/0 | 10.0.2.0/30 | 10.0.2.1 | 10.0.2.2 |

| From | To | Network | From IP | To IP |
|---|---|---|---|---|
| MLS2 Gi0/1 | Core1 Gi0/1 | 10.0.3.0/30 | 10.0.3.1 | 10.0.3.2 |
| MLS2 Gi0/2 | Core2 Gi0/1 | 10.0.4.0/30 | 10.0.4.1 | 10.0.4.2 |
| Core1 Gi0/2 | ISP1 Gi0/0 | 203.0.113.0/30 | 203.0.113.2 | 203.0.113.1 |
| Core2 Gi0/2 | ISP2 Gi0/0 | 203.0.113.4/30 | 203.0.113.6 | 203.0.113.5 |

**Access switch connections**

| Department | Access switch | Connected to |
|---|---|---|
| Sales & Marketing | SW-SALES | MLS1 Fa0/1 |
| HR & Logistics | SW-HR | MLS1 Fa0/2 |
| Finance & Account | SW-FINANCE | MLS1 Fa0/3 |
| Inter-Relation | SW-INTERREL | MLS1 Fa0/4 |
| ICT Engineer | SW-ICT | MLS2 Fa0/1 |
| Server Room | SW-SERVER | MLS2 Fa0/2 |

### 2.3 How the network works

- **Addressing:** a new device asks for an address, the multilayer switch relays the request to Server0, and Server0 answers from the pool of that department.
- **Traffic inside the company:** every device uses its multilayer switch as gateway, and OSPF carries traffic between MLS1, MLS2 and the core routers.
- **Internet traffic:** the default route points to Core1, NAT replaces the private address with 203.0.113.2, and ISP1 sends the reply back.
- **Failure case:** if Core1 or ISP1 fails, Core2 installs its backup default route and traffic uses ISP2 with address 203.0.113.6.

## 3. Configuration

Paste the commands in the CLI of each device in this order: **ISP routers, core routers, multilayer switches, access switches**. The order matters because the uplinks must exist before OSPF neighbors and DHCP relay can work. Full command lists are shown only where they are needed; for repeated devices only the changed values are given.

| Device | Role in the network | Key things configured |
|---|---|---|
| ISP1, ISP2 | Simulated internet | Link IP and a loopback test address |
| Core1, Core2 | Edge routers | OSPF, default route, NAT overload |
| MLS1, MLS2 | Gateways of all departments | Routed ports, DHCP relay, OSPF |
| Access switches | Connect end devices | Hostname only (layer 2) |

The servers and end devices are configured in sections 4 and 5.

### 3.1 ISP1

The ISP routers only simulate the internet. Each needs an IP address on the link to its core router and a loopback address that is used as a ping target.

```
interface GigabitEthernet0/0
 ip address 203.0.113.1 255.255.255.252
 no shutdown
interface Loopback0
 ip address 8.8.8.8 255.255.255.255
```

Set the name with `hostname ISP1` and save with `copy running-config startup-config`.

## 3.2 ISP2

ISP2 uses the same commands as ISP1 with these values (hostname `ISP2`):

| Setting | Value |
|---|---|
| Gi0/0 IP address | 203.0.113.5 255.255.255.252 |
| Loopback0 | 8.8.4.4 255.255.255.255 |
| Connected to | Core2 Gi0/2 |

## 3.3 Core1 (primary path)

Core1 does three jobs: it connects to both multilayer switches, runs OSPF, and translates private addresses to its ISP1 address with NAT.

| Interface | IP address | NAT role | Connected to |
|---|---|---|---|
| Gi0/0 | 10.0.1.2/30 | inside | MLS1 |
| Gi0/1 | 10.0.3.2/30 | inside | MLS2 |
| Gi0/2 | 203.0.113.2/30 | outside | ISP1 |

After setting the IP addresses and `ip nat inside` or `ip nat outside` on these interfaces, the important commands are:

```
ip route 0.0.0.0 0.0.0.0 203.0.113.1
router ospf 1
 router-id 3.3.3.3
 network 10.0.1.0 0.0.0.3 area 0
 network 10.0.3.0 0.0.0.3 area 0
 default-information originate
access-list 1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 interface GigabitEthernet0/2 overload
```

- `ip route 0.0.0.0 ...` sends all unknown traffic to ISP1.
- `default-information originate` shares this default route with MLS1 and MLS2 through OSPF.
- `access-list 1` selects the company addresses (192.168.0.0/16) that may use NAT.
- `overload` (PAT) lets many devices share the single address 203.0.113.2.

## 3.4 Core2 (backup path)

Core2 is configured like Core1, only the values below change. It stays idle for internet traffic until Core1 or ISP1 fails.

| Setting | Core1 (primary) | Core2 (backup) |
|---|---|---|
| Gi0/0 (to MLS1) | 10.0.1.2/30 | 10.0.2.2/30 |
| Gi0/1 (to MLS2) | 10.0.3.2/30 | 10.0.4.2/30 |

| Setting | Core1 (primary) | Core2 (backup) |
|---|---|---|
| Gi0/2 (to ISP) | 203.0.113.2/30 | 203.0.113.6/30 |
| OSPF router-id | 3.3.3.3 | 4.4.4.4 |
| Default route | next hop 203.0.113.1 | next hop 203.0.113.5, distance 200 |
| Default route in OSPF | normal metric | metric 100 |

The two lines that create the backup behaviour are:

```
ip route 0.0.0.0 0.0.0.0 203.0.113.5 200
default-information originate metric 100
```

The second line is typed inside `router ospf 1`. The distance 200 makes the route a floating static route: it is used only when the better default route learned from Core1 disappears. NAT (`ip nat inside`, `ip nat outside`, access-list 1 and the overload command) is configured exactly as on Core1, with Gi0/2 as the outside interface.

## 3.5 MLS1 (Building-1 and Building-2)

MLS1 is the default gateway for four departments, so each department port becomes a routed port (`no switchport`) that carries the gateway IP of that department. Turn on routing first with `ip routing`.

Example for Sales & Marketing (port Fa0/1):

```
ip routing
interface FastEthernet0/1
 no switchport
 ip address 192.168.10.1 255.255.255.192
 ip helper-address 192.168.12.194
 no shutdown
```

Repeat the same pattern on the other ports with these values:

| Port | Connected to | IP address (gateway) | Mask |
|---|---|---|---|
| Fa0/1 | SW-SALES | 192.168.10.1 | 255.255.255.192 |
| Fa0/2 | SW-HR | 192.168.10.65 | 255.255.255.192 |
| Fa0/3 | SW-FINANCE | 192.168.10.129 | 255.255.255.192 |
| Fa0/4 | SW-INTERREL | 192.168.10.193 | 255.255.255.192 |
| Gi0/1 | Core1 | 10.0.1.1 | 255.255.255.252 |
| Gi0/2 | Core2 | 10.0.2.1 | 255.255.255.252 |

The Gi0/1 and Gi0/2 ports do not need `ip helper-address`.

**How DHCP relay works:** a PC sends a broadcast to find a DHCP server, but broadcasts do not cross routers and the server is in another subnet. With `ip helper-address`, MLS1 turns the broadcast into a unicast message to Server0 (192.168.12.194) and adds the department gateway address. Server0 uses that address to choose the matching pool and replies through MLS1 to the PC.

Then configure OSPF on MLS1:

```
router ospf 1
 router-id 1.1.1.1
 network 192.168.10.0 0.0.0.255 area 0
```

```
 network 10.0.1.0 0.0.0.3 area 0
 network 10.0.2.0 0.0.0.3 area 0
 passive-interface FastEthernet0/1
 passive-interface FastEthernet0/2
 passive-interface FastEthernet0/3
 passive-interface FastEthernet0/4
```

One `network` line with wildcard 0.0.0.255 covers all four department /26 subnets. `passive-interface` stops OSPF messages from going towards the users, but the subnets are still advertised to the rest of the network. Finish with `copy running-config startup-config`.

### 3.6 MLS2 (Building-3)

MLS2 is the gateway for ICT Engineer and the Server Room. The Server Room needs no helper address because the DHCP server is in the same subnet. Use the same pattern as MLS1 (`ip routing`, `no switchport`, IP address, `no shutdown`) with these values:

| Port | Connected to | IP address | Mask | DHCP relay |
|---|---|---|---|---|
| Fa0/1 | SW-ICT | 192.168.11.1 | 255.255.255.192 | Yes, 192.168.12.194 |
| Fa0/2 | SW-SERVER | 192.168.12.193 | 255.255.255.248 | No (server is in this subnet) |
| Gi0/1 | Core1 | 10.0.3.1 | 255.255.255.252 | No |
| Gi0/2 | Core2 | 10.0.4.1 | 255.255.255.252 | No |

OSPF configuration:

```
router ospf 1
 router-id 2.2.2.2
 network 192.168.11.0 0.0.0.255 area 0
 network 192.168.12.192 0.0.0.7 area 0
 network 10.0.3.0 0.0.0.3 area 0
 network 10.0.4.0 0.0.0.3 area 0
 passive-interface FastEthernet0/1
 passive-interface FastEthernet0/2
```

The wildcard mask 0.0.0.7 matches the /29 Server Room subnet. Save with `copy running-config startup-config`.

### 3.7 Access switches

The access switches only work at layer 2 with the default VLAN, so they only need a name. Repeat the commands on every access switch and change the hostname: SW-SALES, SW-HR, SW-FINANCE, SW-INTERREL, SW-ICT, SW-SERVER.

```
hostname SW-SALES
no ip domain-lookup
line console 0
 logging synchronous
```

`no ip domain-lookup` stops the switch from waiting when a command is mistyped, and `logging synchronous` keeps log messages from cutting into the command you are typing.

## 4. Server Configuration

The servers are configured from the GUI. Click the server, open the tab and fill in the values below.

## 4.1 IP settings of the servers

Go to **Desktop tab > IP Configuration > Static**.

| Server | IP address | Subnet mask | Default gateway | DNS server |
|---|---|---|---|---|
| Server0 (DHCP) | 192.168.12.194 | 255.255.255.248 | 192.168.12.193 | 192.168.12.197 |
| Server2 (HTTP) | 192.168.12.196 | 255.255.255.248 | 192.168.12.193 | 192.168.12.197 |
| Server1 (DNS) | 192.168.12.197 | 255.255.255.248 | 192.168.12.193 | 192.168.12.197 |

**Figure 2:** IP configuration of Server0, Server1 and Server2 *(insert screenshot here)*

## 4.2 DHCP server (Server0)

Go to **Services tab > DHCP** and set Service to **On**. Delete the default pool or leave it unused, then add one pool for each department with the values below. Click **Add** after every pool.

| Pool name | Default gateway | DNS server | Start IP address | Subnet mask | Max users |
|---|---|---|---|---|---|
| SALES_MARKETING | 192.168.10.1 | 192.168.12.197 | 192.168.10.10 | 255.255.255.192 | 50 |
| HR_LOGISTICS | 192.168.10.65 | 192.168.12.197 | 192.168.10.74 | 255.255.255.192 | 50 |
| FINANCE_ACCOUNT | 192.168.10.129 | 192.168.12.197 | 192.168.10.138 | 255.255.255.192 | 50 |
| INTER_RELATION | 192.168.10.193 | 192.168.12.197 | 192.168.10.202 | 255.255.255.192 | 50 |
| ICT_ENGINEER | 192.168.11.1 | 192.168.12.197 | 192.168.11.10 | 255.255.255.192 | 50 |

**Figure 3:** DHCP service of Server0 with all five pools *(insert screenshot here)*

## 4.3 DNS server (Server1)

Go to **Services tab > DNS**, set DNS Service to **On**, then add this record:

| Name | Type | Address |
|---|---|---|
| www.company.com | A Record | 192.168.12.196 |

**Figure 4:** DNS service of Server1 with the A record *(insert screenshot here)*

## 4.4 HTTP server (Server2)

Go to **Services tab > HTTP** and make sure HTTP is **On**. The default `index.html` can be edited so the page shows the company name.

**Figure 5:** HTTP service of Server2 *(insert screenshot here)*

# 5. End Device Configuration

All PCs, laptops, printers, tablets and smartphones are set to DHCP.

- **Wired devices:** Desktop tab > IP Configuration > select **DHCP**.
- **Wireless devices:** open the PC Wireless / Config tab, connect to the SSID of the access point of that department and select **DHCP**.

**Figure 6:** A PC after getting its IP address from DHCP (`ipconfig`) *(insert screenshot here)*

# 6. Verification and Testing

## 6.1 Check interfaces and routing

Run these commands on the multilayer switches and routers.

| Command | What it shows | What to look for |
|---|---|---|
| `show ip interface brief` | Interface IPs and status | Status and protocol both up |
| `show ip route` | Routing table | OSPF routes (O) and a default route (O*E2) |
| `show ip ospf neighbor` | OSPF neighbors | State FULL with every neighbor |
| `show running-config` | Full configuration | Matches the settings in section 3 |

**Figure 7:** `show ip interface brief` on MLS1 *(insert screenshot here)*

**Figure 8:** `show ip ospf neighbor` on Core1 *(insert screenshot here)*

**Figure 9:** `show ip route` on MLS1, OSPF routes and default route *(insert screenshot here)*

## 6.2 Check DHCP on the end devices

On a PC open the Command Prompt and run:

```
ipconfig /all
```

The PC should show an address from its department range, the correct gateway and the DNS server 192.168.12.197.

**Figure 10:** `ipconfig /all` on a PC of each department *(insert screenshot here)*

## 6.3 Connectivity tests

From a PC in Sales & Marketing, run the following tests from the Command Prompt:

| Test | What it checks | Expected result |
|---|---|---|
| `ping 192.168.10.1` | Own default gateway (MLS1) | Reply received |
| `ping 192.168.12.194` | Reach the DHCP server in the Server Room | Reply received |
| `ping 192.168.10.74` (or any HR device) | Communication between departments | Reply received |
| `ping 192.168.11.1` | Reach the ICT department gateway (MLS2) through the core | Reply received |
| `ping 8.8.8.8` | Internet through Core1, NAT and ISP1 | Reply received |

**Figure 11:** Successful ping to the gateway and the DHCP server *(insert screenshot here)*

**Figure 12:** Successful ping between two different departments *(insert screenshot here)*

**Figure 13:** Successful ping to 8.8.8.8 (internet) *(insert screenshot here)*

## 6.4 DNS and web server test

On a PC open the Web Browser and type `http://www.company.com`. The web page of Server2 should open. This confirms that DNS and HTTP are both working.

```
ping www.company.com
```

**Figure 14:** Web browser showing the page from www.company.com *(insert screenshot here)*

### 6.5 NAT check

After pinging 8.8.8.8, run this on Core1:

```
show ip nat translations
show ip nat statistics
```

**Figure 15:** `show ip nat translations` on Core1 *(insert screenshot here)*

### 6.6 Failover test (backup ISP)

To prove the backup path works:

1. On Core1, shut down the link to ISP1: `interface GigabitEthernet0/2` then `shutdown`.
2. Wait a few seconds for OSPF to remove the default route learned from Core1.
3. From a PC, run `ping 8.8.4.4`. Traffic should now go through Core2 and ISP2.
4. Run `show ip nat translations` on Core2 to confirm translations are created there.
5. Bring the link back with `no shutdown` on Core1.

**Figure 16:** Ping to 8.8.4.4 and NAT translations on Core2 during failover *(insert screenshot here)*

## 7. Troubleshooting Guide

| Problem | Possible cause | What to check / fix |
|---|---|---|
| PC gets 169.254.x.x address | DHCP not reaching the server | Check `ip helper-address` on the department port, and that Server0 DHCP service is On |
| PC gets address but no gateway reply | Wrong pool gateway | Compare the pool gateway with the MLS interface IP |
| No OSPF neighbors | Interface down or network statement missing | `show ip interface brief`, then check `network` commands and area 0 |
| Cannot reach other departments | Missing OSPF network or passive interface mistake | `show ip route` on MLS1 and MLS2 |
| Cannot ping 8.8.8.8 | NAT or default route problem | `show ip nat statistics` and `show ip route 0.0.0.0` on Core1 |
| Website does not open by name | DNS record missing | Verify the A record on Server1 and the DNS server address in the DHCP pool |
| Wireless device not getting IP | Not connected to the right SSID | Reconnect to the department access point and select DHCP |

## 8. Conclusion

In this project a complete company network was designed and implemented in Cisco Packet Tracer. Every department has its own subnet, devices get their addresses automatically from the central DHCP server through DHCP relay, OSPF connects all the subnets, and NAT with two ISPs gives internet access with a backup path. The DNS and HTTP servers in the Server Room are reachable from all departments.

## 8.1 Future scope

- Add VLANs and trunking on the access switches for better traffic separation.
- Add ACLs to control access between departments and to the Server Room.
- Enable port security on access switches and secure the CLI with passwords and SSH.
- Add HSRP/VRRP between the multilayer switches for gateway redundancy.
- Add a wireless controller or WPA2 security on the access points.
- Add IPv6 addressing and monitoring using SNMP or Syslog.

# Appendix: Abbreviations

| Term | Meaning |
|---|---|
| DHCP | Dynamic Host Configuration Protocol |
| DNS | Domain Name System |
| HTTP | Hypertext Transfer Protocol |
| OSPF | Open Shortest Path First |
| NAT / PAT | Network Address Translation / Port Address Translation (overload) |
| ISP | Internet Service Provider |
| MLS | Multilayer Switch |
| SSID | Service Set Identifier (wireless network name) |